Initialization ($t = 1$) **Phase Transition ($t = \tau T$)** **Termination ($t = T$)**

Phase 1: Sparse Subset Exploration

$$f(\mathbf{x}, \mathbf{w}) = w_{\text{acc}} \cdot \text{Error}(\mathbf{x}) + w_{\text{feat}} \cdot \frac{\|\mathbf{x}\|_0}{D}$$

High Feature-Ratio Penalty ($w_{\text{feat}} > w_{\text{acc}}$)

Sparsity-oriented exploration · Progressive feature reduction

Adaptive Weight Scheduling

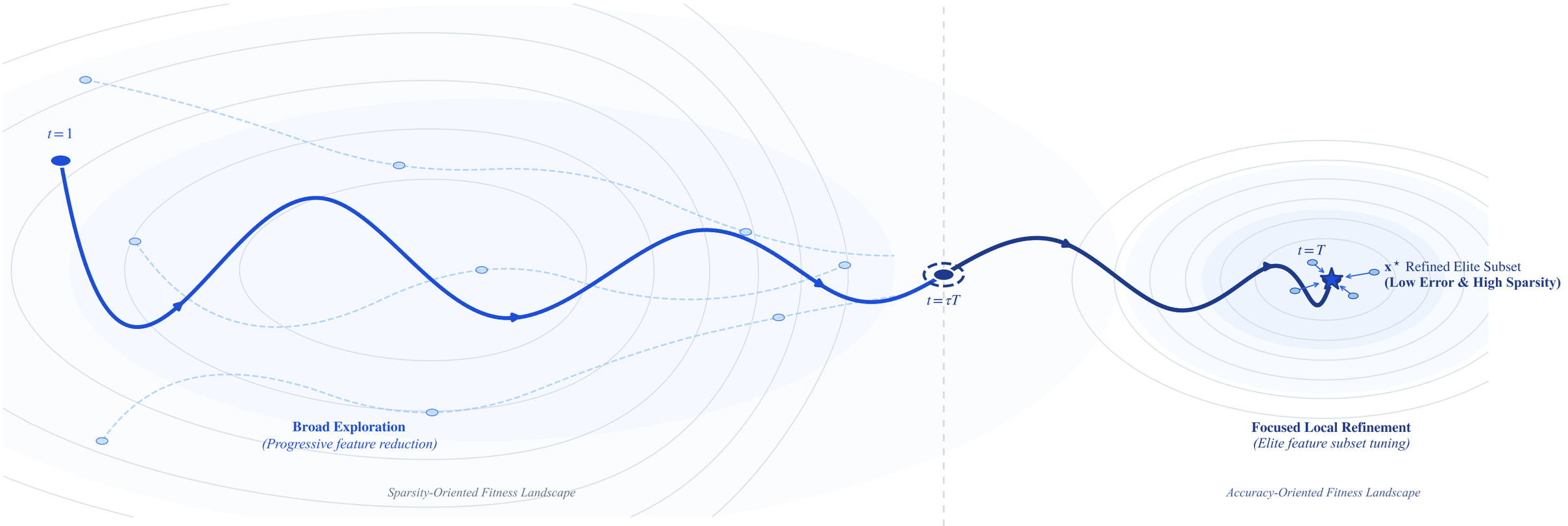
$$\mathbf{w}_{\text{Phase 1}} \longrightarrow \mathbf{w}_{\text{Phase 2}}$$

Phase 2: Accuracy Refinement

$$f(\mathbf{x}, \mathbf{w}) = w_{\text{acc}} \cdot \text{Error}(\mathbf{x}) + w_{\text{feat}} \cdot \frac{\|\mathbf{x}\|_0}{D}$$

Classification Fidelity Priority ($w_{\text{acc}} \gg w_{\text{feat}}$)

Accuracy-oriented refinement · Elite subset fine-tuning



Exploration → **Feature Reduction** → **Weight Transition** → **Accuracy Refinement** → **High-Quality Compact Subset**